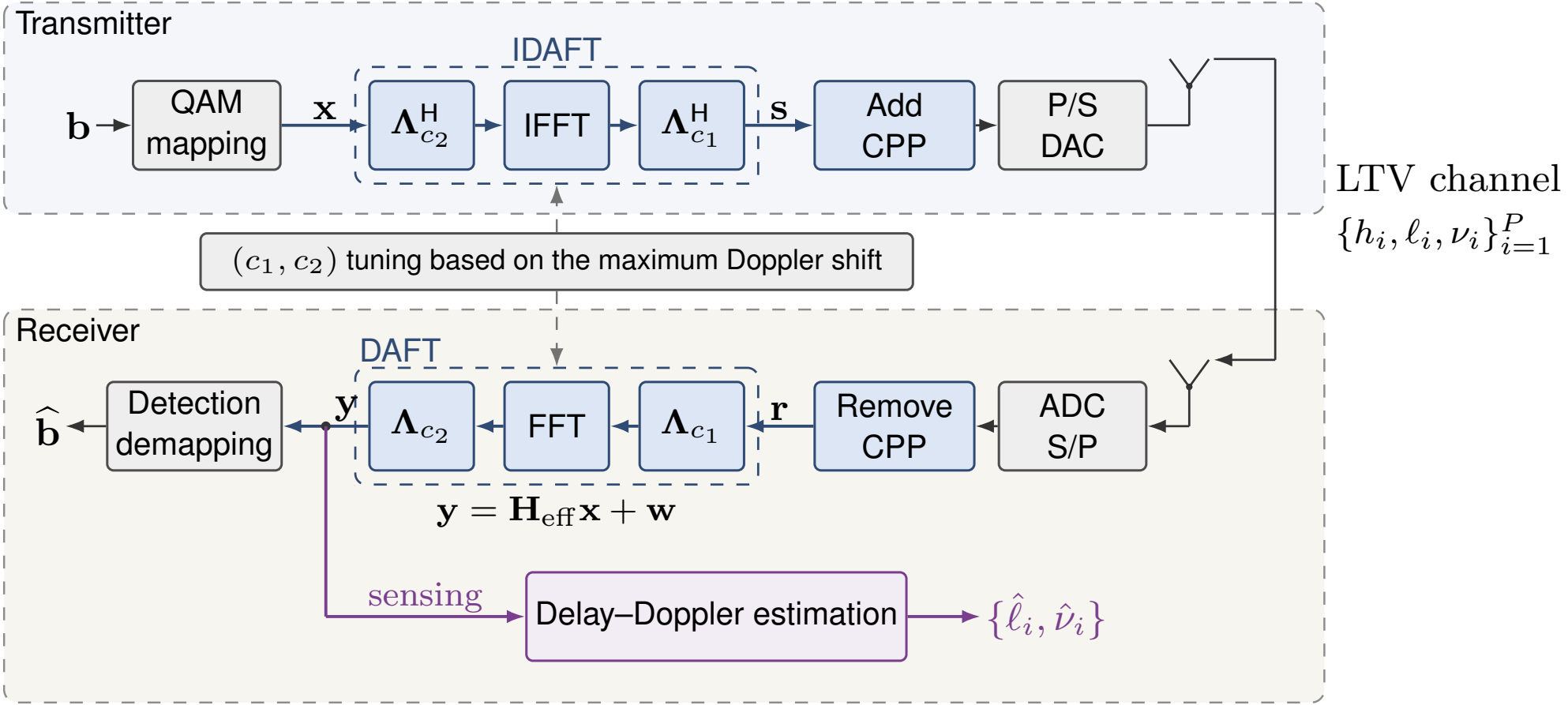

Transmitter
IDAFT
b
QAM mapping
x
$\mathbf{\Lambda}_{c_2}^{\mathrm{H}}$
IFFT
$\mathbf{\Lambda}_{c_1}^{\mathrm{H}}$
s
Add CPP
P/S DAC
LTV channel
$\{h_i, \ell_i, \nu_i\}_{i=1}^{P}$
(c_1, c_2) tuning based on the maximum Doppler shift
Receiver
DAFT
$\widehat{\mathbf{b}}$
Detection demapping
y
$\mathbf{\Lambda}_{c_2}$
FFT
$\mathbf{\Lambda}_{c_1}$
r
Remove CPP
ADC S/P
$\mathbf{y} = \mathbf{H}_{\mathrm{eff}}\mathbf{x} + \mathbf{w}$
sensing
Delay–Doppler estimation
$\{\hat{\ell}_i, \hat{\nu}_i\}$